

Project Report

Github URL: <https://github.com/harshithaattantiwebspace/AutoCorp-Hub>

Video URL: <https://youtu.be/u1ALsFHpEmk>

1. Title and List of Project Participants:

Project Title: AutoCorp Hub : AI-Powered Multi-Agent Platform for Enterprise Automation

Project Participants:

1. Harshitha Attanti
2. Manohar Korikana

2. Project Goals:

AutoCorp Hub's goal is to build a cloud-based AI platform where businesses can hire digital workers (AI agents) to run their daily business tasks.

A shared orchestrator connects all of the agents (like HR Onboarding, Mail Automation, Meeting Scheduler, and Jira Manager) into a single microservice.

Key objectives:

- Automate HR onboarding workflows (email parsing, calendar scheduling, and document generation).
- Build and deploy scalable microservices using **Google Kubernetes Engine (GKE)**.
- Use **Gmail API** for parsing onboarding or interview requests and triggering downstream workflows.
- Use **Google Calendar API** to automatically schedule and manage HR events and meetings.

- Develop a **Streamlit-based dashboard** for HR managers to monitor automation and control workflows.
- Integrate **LLM models** (through Hugging Face or Vertex AI) to interpret natural-language email content and extract structured onboarding data.
- Deploy, monitor, and scale the platform entirely within **Google Cloud Platform**.

3. Specific List of software and Hardware Components:

Software Components:

Component	Description
Frontend: Streamlit	Interface for subscribing to the micro agents and use them.
Backend Framework: FastAPI	REST microservices managing onboarding workflows, Gmail/Calender integration.
LLM Engine: Hugging Face API/Vertex AI	Process EMAIL content
Containerization: Docker	Packages each microservice for deployment on GKE.
Orchestration: GKE	Manages microservices , scaling, and networking.
Authentication: Google OAuth2	Used for secure access to Gmail and calender.
Storage: Google Cloud Storage(GCS)	Stores onboarding forms, resumes, and generated offer letters.

Hardware Components:

Resource	Description
Developer Machines	Local Laptops for development and testing

GCP Compute Nodes	Virtual machines for hosting GKE clusters.
GPU Node	For faster LLM inference and text extraction.

4. Architectural Diagram:

Streamlit UI: This acts as the central dashboard for HR and employees to activate, monitor, and control automation agents through interactive web interface.

Agent Service: Handles the API requests from streamlit UI and routes them to the appropriate backend microservice.

Microservices: Set of independent service modules that performs particular automation task.

Mail Automation: Automatically reads the incoming emails, classifies them and triggers relevant workflows using Gmail API.

Meeting Scheduler: This detects requests from emails and book slots in google calendar, even avoiding scheduling conflicts.

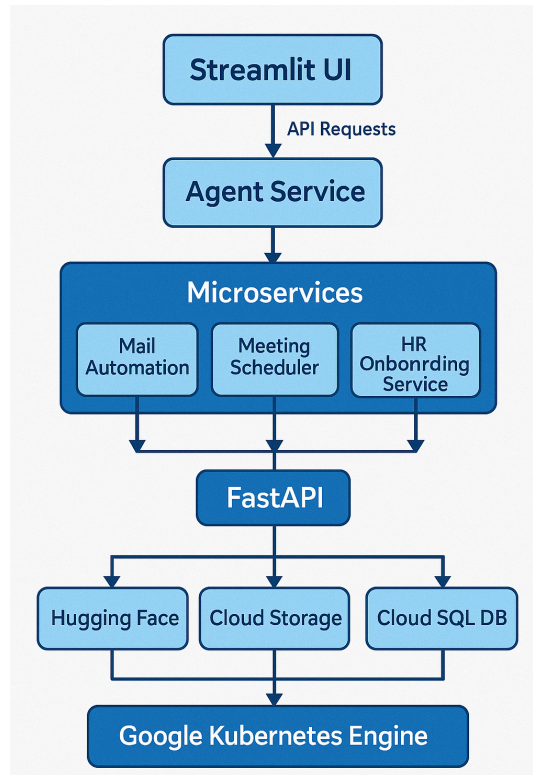
HR Onboarding Service: This fetches employee data from cloud SQL, retrieves required documents from buckets, and responds with onboarding materials.

FastAPI: Serves as backend framework managing communication between microservices, APIs and cloud integrations.

Cloud Storage: Stores all HR-related documents such as resumes, offer letters, and onboarding forms for access and retrieval.

Cloud SQL DB: Maintains structured employee records and system metadata used by microservices for decision making.

Google Kubernetes Engine(GKE): Provides orchestration, scaling, and deployment infrastructure for all containerized services.



5. Interaction between Software and Hardware Components:

- **User Interaction(Streamlit UI):** The HR manager safely logs into the system with Google OAuth2.

The user can choose agent services like Mail automation, Meeting scheduler, and HR Onboarding Service from the dashboard.

The Agent service sends an API request to the backend when the user starts an automation.

- **Agent Service:** The streamlit UI sends API requests to the agent service. It figures out which microservice should do the job based on what the user chose.
- **Microservices Layer(Mail Automation, Meeting Scheduler, HR Onboarding):** uses the Gmail API to read emails and get the subject, body, attachments, and sender information. LLM for getting structured information out of data.

Use the Calender API to automatically set up meetings or interviews based on information taken from emails. Sends calendar invites to people who are going to the event and changes the schedules on the fly.

Makes onboarding documents like offer letters, welcome emails, and HR forms. Uploads files that were made to GCS.

- **Backend Orchestration (FastAPI):** FastAPI acts as main orchestration and routing engine for all microservices.
- **Infrastructure Orchestration(Google Kubernetes Engine):** All components are deployed as Docker containers within GKE.

6. Why choose this component?

Component	Why it was Chosen	Advantages	Disadvantages
Streamlit(Frontend UI)	Simple and fast way to build interactive dashboard	Easy integration with python scripts	Not suitable for large scale production systems.
FastAPI (Backend Framework)	Asynchronous API layer to manage requests from multiple agents.	High performance async IO, easy to integrate with databases and external APIs.	Less built in admin tools.
Docker (containerization)	To Package each service to ensure consistent deployment on any environment.	Simplifies deployment and scaling	Increases image size and requires proper version management.
Google Kubernetes Engine (GKE)	For orchestrating multiple agent containers and scaling them.	Automatic scaling, load balancing.	Slightly complex setup, higher costs for small-scale projects.

Google OAuth2 (Authentication)	For secure authentication for accessing Gmail and calendar APIs.	Secure, seamless integration with google APIs.	Tokens expire periodically and require refresh logic.
Google Cloud SQL	To manage postgresSQL database for employee and HR data	Fully managed, integrated with GCP services.	Limited customization.
Google cloud storage	To store resumes, HR documents securely.	Highly scalable and reliable storage, easy API access.	Slightly higher latency compared to local storage.

7. Debugging, training and Testing Mechanisms:

- **Debugging Approach:**

Use FastAPI to independently test every microservice. To track errors, use Python logging. To track inter-service latency and locate bottlenecks, turn on Google Cloud Trace.

- **Testing & Validation:**

Pytest for API endpoints, LLM extraction functions, and DB interaction.

Integration Tests: Validate end-to-end flow from Gmail->LLM->Calendar->HR document generation.

Load Testing: Use Locust to simulate concurrent HR users performing automation.

LLM Validation: use a dataset of HR emails to evaluate extraction accuracy and model reliability.

8. Working System and Potential Bottlenecks:

Working System:

AutoCorp Hub system enables automated operations through an integrated set of AI agents. User interacts with the streamlit dashboard, which communicates with the Agent service via REST APIs. These requests are routed to their

dedicated microservices such as Mail automation, Meeting scheduler, and HR Onboarding where each will perform its specific task.

The backend services, built with FastAPI, process requests efficiently by interacting with Google Cloud SQL for employee data and Google cloud storage for HR files. All services are containerized using docker and orchestrated through Google Kubernetes Engine (GKE), which enables load balancing, scaling.

Expected Workload Capacity:

Traffic load: The system can manage multiple automation requests across agents.

Response Time: The architecture is optimized for sub-second responses for most API calls, with caching mechanisms for frequently accessed employee and document data.

Reliability: Each microservice operates independently, ensuring that failure in one component does not affect others.

Potential Bottlenecks:

Database Query Latency: High-volume SQL queries in Cloud SQL may slow down performance during peak load.

Email parsing and LLM inference: Processing very large volumes of incoming emails with LLMs could introduce latency.

Cloud Storage Access: Fetching or uploading large HR documents to GCS may delay responses under heavy load.

Scaling during concurrent requests: Extreme traffic surges may temporarily overload the system.

Authentication Token Expiry: Frequent API interactions with Gmail and Calendar may fail if tokens expire.

Conclusion:

The AutoCorp Hub system is a scalable AI-driven platform to automate enterprise operations, particularly within HR departments. It integrates multiple intelligent agents such as Mail automation, meeting scheduler, HR onboarding into a common cloud based microservice architecture.

This application is built on Google Cloud Platform and orchestrated through Google Kubernetes Engine(GKE), Autocorp ensures scalability, reliability. Overall, the platform provides a practical demonstration of how agentic AI and cloud native design can transform traditional enterprise processes into autonomous workflows.

Future Scope:

1. Integration with additional agents.
2. Adding advanced LLM capabilities.
3. Voice and chat integration to the current platform.
4. Cross-cloud and API expansion.
5. Enable continuous learning Agents.